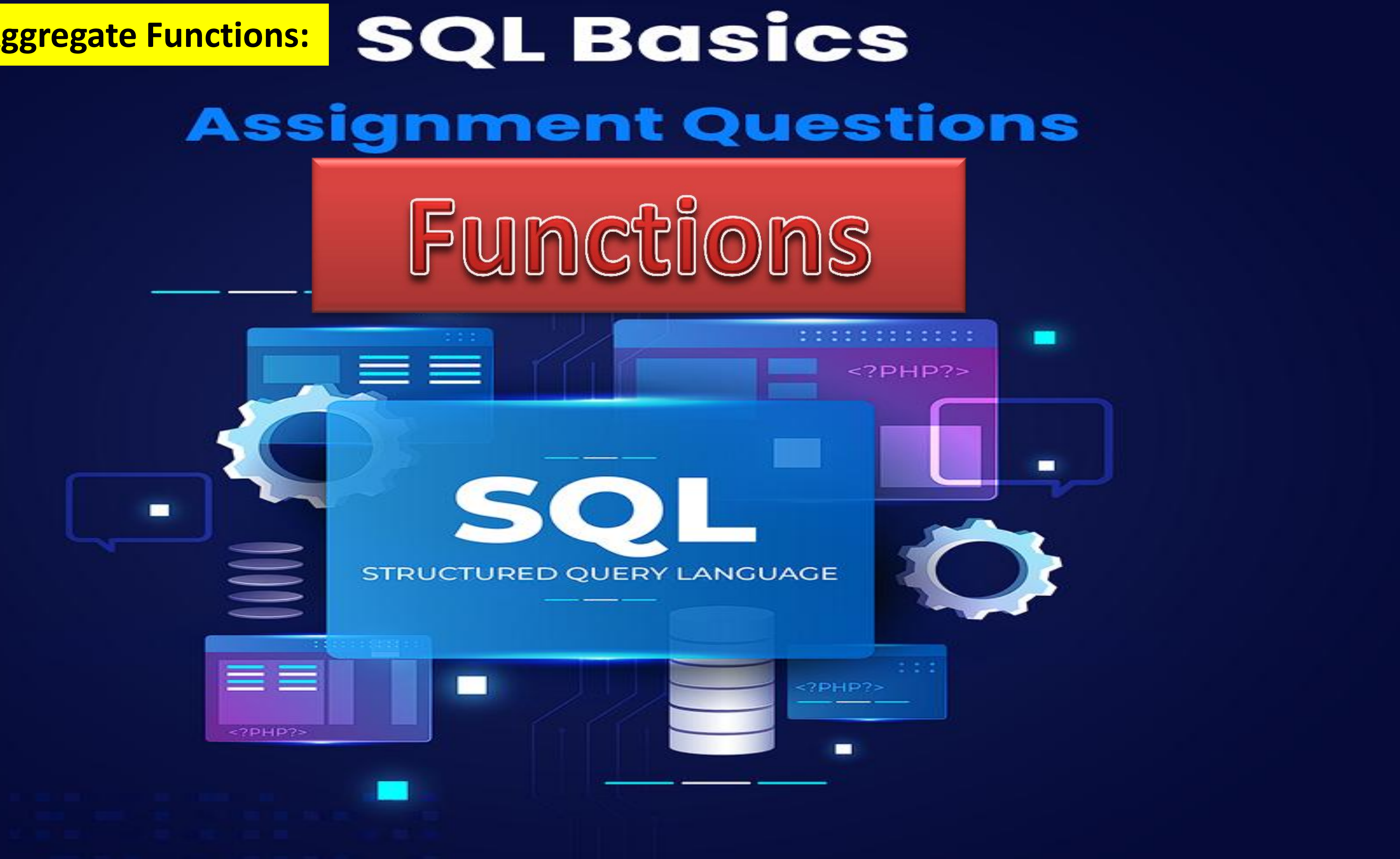
Basic Aggregate Functions:



1

- Retrieve the total number of rentals made in the Sakila database.
  Hint: Use the COUNT() function.

Input

```sql
SELECT COUNT(*) AS total_rentals
FROM rental;
```

Result Grid

total_rentals
16044

Output

2

- Find the average rental duration (in days) of movies rented from the Sakila database. Hint: Utilize the AVG() function.

Input

```sql
SELECT AVG(DATEDIFF(return_date, rental_date)) AS avg_rental_duration
FROM rental;
```

Result Grid | Filter Rows:

avg_rental_duration
5.0252

Output

3

- Display the first name and last name of customers in uppercase. Hint: Use the UPPER () function..

Input

```sql
SELECT UPPER(first_name) AS upper_first_name,
       UPPER(last_name) AS upper_last_name
FROM customer;
```

Output

Result Grid | Filter Rows:

upper_first_name	upper_last_name
MARY	SMITH
PATRICIA	JOHNSON
LINDA	WILLIAMS
BARBARA	JONES
ELIZABETH	BROWN
JENNIFER	DAVIS
MARIA	MILLER
SUSAN	WILSON
MARGARET	MOORE
DOROTHY	TAYLOR
LISA	ANDERSON
NANCY	THOMAS
KAREN	JACKSON
BETTY	WHITE
HELEN	HARRIS
SANDRA	MARTIN
DONNA	THOMPSON
CAROL	GARCIA
RUTH	MARTINEZ

4

- Extract the month from the rental date and display it alongside the rental ID. Hint: Employ the MONTH() function.

Input

```sql
SELECT rental_id,
       MONTH(rental_date) AS rental_month
FROM rental;
```

Output

Result Grid | Filter Rows

rental_id	rental_month
1	5
2	5
3	5
4	5
5	5
6	5
7	5
8	5
9	5
10	5
11	5
12	5
13	5
14	5
15	5
16	5
17	5
18	5

5

- Retrieve the count of rentals for each customer (display customer ID and the count of rentals). Hint: Use COUNT () in conjunction with GROUP BY.

Input

```sql
SELECT customer_id,
       COUNT(*) AS rental_count
FROM rental
GROUP BY customer_id;
```

Output

Result Grid | Filter Rows:

customer_id	rental_count
1	32
2	27
3	26
4	22
5	38
6	28
7	33
8	24
9	23
10	25
11	24
12	28
13	27
14	28
15	32
16	28
17	21

6

- Find the total revenue generated by each store. Hint: Combine SUM() and GROUP BY.

Input

```
SELECT s.store_id,
       SUM(p.amount) AS total_revenue
FROM payment p
JOIN staff s ON p.staff_id = s.staff_id
GROUP BY s.store_id;
```

Output

Result Grid | Filter Ro

store_id	total_revenue
1	33489.47
2	33927.04

7

- Determine the total number of rentals for each category of movies.
  Hint: JOIN film_category, film, and rental tables, then use COUNT () and GROUP BY.

Input

```sql
SELECT c.name AS category_name,
       COUNT(r.rental_id) AS total_rentals
FROM category c
JOIN film_category fc ON c.category_id = fc.category_id
JOIN film f ON fc.film_id = f.film_id
JOIN inventory i ON f.film_id = i.film_id
JOIN rental r ON i.inventory_id = r.inventory_id
GROUP BY c.name;
```

Result Grid | Filter Rows:

category_name	total_rentals
Action	1112
Animation	1166
Children	945
Classics	939
Comedy	941
Documentary	1050
Drama	1060
Family	1096
Foreign	1033
Games	969
Horror	846
Music	830
New	940
Sci-Fi	1101
Sports	1179
Travel	837

Output

8

- Find the average rental rate of movies in each language. Hint: JOIN film and language tables, then use AVG () and GROUP BY.

Input

```sql
SELECT l.name AS language,
       AVG(f.rental_rate) AS avg_rental_rate
FROM film f
JOIN language l ON f.language_id = l.language_id
GROUP BY l.name;
```

Result Grid   Filter Rows:

language	avg_rental_rate
English	2.980000

Output

THANKS FOR THIS PROJECT